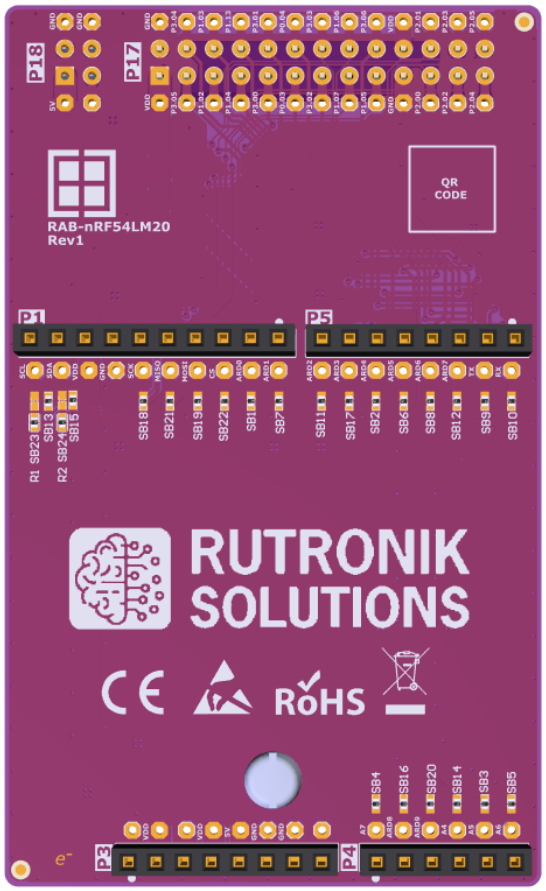
RAB – nRF54LM20

User Manual



Versions

Version	Date	Rationale
0.1	September 14, 2026	First draft. Author: GDR
1.0	October 30, 2026	First release. Author: KOA

Legal Disclaimer

The evaluation board is for testing purposes only and, because it has limited functions and limited resilience, is not suitable for permanent use under real conditions. If the evaluation board is nevertheless used under real conditions, this is done at one's own responsibility; any liability of Rutronik is insofar excluded.

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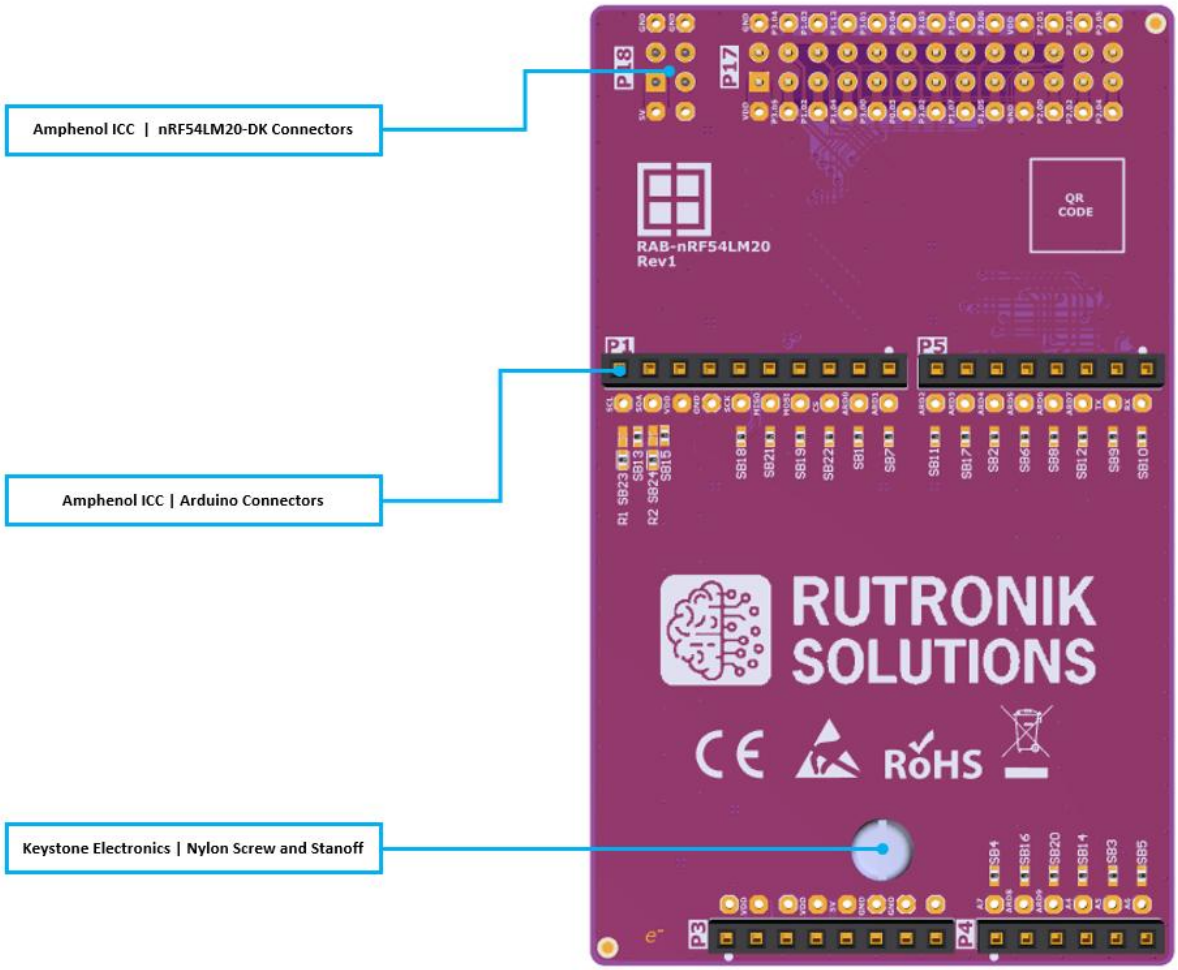
Overview

This simple but useful adapter board lets you use the Nordic nRF54LM20-DK development kit with almost all Rutronik System Solutions adapter boards. The nRF54LM20-DK does not have Arduino pin headers; therefore, Rutronik System Solutions developed this bridge-adapter board to support Rutronik´s and third-party adapter boards that require an Arduino interface.

Features

- **Arduino Connectors** – Amphenol ICC 75915-310LF, 75915-308LF, 75915-306LF.
- **nRF54LM20-DK Connectors -** Amphenol ICC 87606-313LF, 87606-302LF.
- **M3 Nylon screw and Standoff -** Keystone Electronics 29341, 25512.
- **Passive components** – Passive components from Yageo.

Component Placement

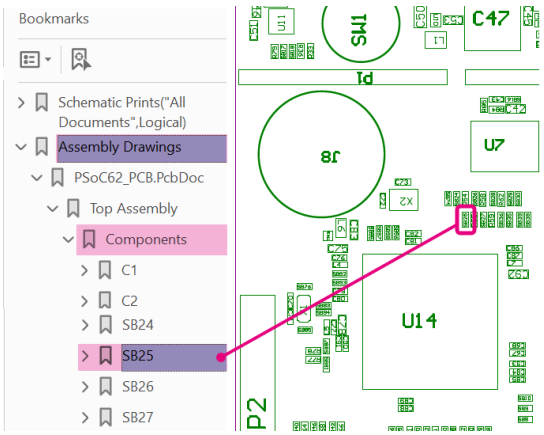


Hardware

Solder Bridges

Designator	Circuit	Default
SB1	P3.05 signal to Arduino IO0	Closed
SB2	P3.04 signal to Arduino IO4	Closed
SB3	P1.02 signal to Arduino A5	Closed
SB4	P1.03 signal to Arduino A7	Closed
SB5	P1.04 signal to Arduino A6	Closed
SB6	P1.13 signal to Arduino IO5	Closed
SB7	P3.00 signal to Arduino IO1	Closed
SB8	P3.01 signal to Arduino IO6	Closed
SB9	P0.03 signal to Arduino TX (UART)	Closed
SB10	P0.04 signal to Arduino RX (UART)	Closed
SB11	P3.02 signal to Arduino IO1	Closed
SB12	P3.03 signal to Arduino IO7	Closed
SB13	P1.07 signal to Arduino SCL (I2C)	Closed
SB14	P1.06 signal to Arduino A4	Closed
SB15	P1.05 signal to Arduino SDA (I2C)	Closed
SB16	P3.06 signal to Arduino IO8	Closed
SB17	P2.00A signal to Arduino IO3	Closed
SB18	P2.01A signal to Arduino SCK (SPI)	Closed
SB19	P2.02A signal to Arduino MOSI (SPI)	Closed
SB20	P2.03A signal to Arduino IO9	Closed
SB21	P2.04A signal to Arduino MISO (SPI)	Closed
SB22	P2.05A signal to Arduino CS (SPI)	Closed
SB23	SCL pull-up resistor	Opened
SB24	SDA pull-up resistor	Opened

You can find the solder bridge locations in the 3D model and assembly drawings of RAB-nRF54LM20.



How to find a component on the layout

Changing the Solder Bridges

The SMD „Chipping Tool - Tweezers“ is recommended for use for SMD solder bridges or fuse soldering on the RAB-nRF54LM20 development board.

RSS Adapter Boards Compatibility Table

Connector		Arduino	RAB1-Sensorfusion	RAB2-CO2	RAB3-Radar	RAB4-RTK	RAB6-Vishay	RAB7-Sensorfusion
P17								
1	VDDIO	3.3V	3.3V	3.3V	VREF	3.3V ARDUINO	3.3V_ARD	3.3V
2	GND	GND	GND	GND	GND	GND	GND	GND
3	P3.05	ARD_IO0			GPIO0_3V3	GNSS_ERR	ARD_IO1	SPI_CS_BME
4	P3.04	ARD_IO4		PSEL	MOD_RST	ON_OFF_CTRL		BMI_INT2
5	P1.02	AN_A5						
6	P1.03	AN_A7				LTE_TX		
7	P1.04	AN_A6				PWRMON_3V3		
8	P1.13	ARD_IO5	SENS_INT2	SENS_INT	IRQ_VREF	PWR_LTE_N		BMM_INT
9	P3.00	ARD_IO1	SENS_INT1	SENS_INT	GPIO1_3V3	GNSS_STA		SPI_CS_BMP
10	P3.01	ARD_IO6			MOD_PWR	PWR_GNSS_N		BMP_INT
11	P0.03	UART_TX			RX_VREF	GNSS_RX1		
12	P0.04	UART_RX			TX_VREF	GNSS_TX1		
13	P3.02	ARD_IO2			SEN_PWR	GNSS_PVT		SPI_CS_DPS
14	P3.03	ARD_IO7		SHDN		GNSS_RST	ARD_IO8	LDO_EN
15	P1.07	I2C_SCL	I2C_SCL	ARD_SCL			I2C_SCL	I2C_SCL
16	P1.06	AN_A4	SENS_INT3			LTE_RTS	AN_INT	
17	P1.05	I2C_SDA	I2C_SDA	ARD_SDA			I2C_SDA	I2C_SDA
18	P3.06	ARD_IO8				LTE_RX		
19	GND	GND						
20	VDDIO	3.3V						
21	P2.00_A	ARD_IO3		LED	SEN_RST_VREF	SHDN_CTRL		BMI_INT1
22	P2.01_A	SPI_SCK	SPI_SCK		SCK_VREF	LTE_CLK		SPI_SCK
23	P2.02_A	SPI_MOSI	SPI_MOSI		MOSI_VREF	LTE_MOSI		SPI_MOSI
24	P2.03_A	ARD_IO9				LTE_CTS		
25	P2.04_A	SPI_MISO	SPI_MISO		MISO_VREF	LTE_MISO		SPI_MISO
26	P2.05_A	SPI_CS	SPI_CS		CS_VREF	LTE_CS		SPI_CS_BMI
P18								
1	5V				5V_ARDUINO	5V_ARDUINO	5V_ARD	
2	GND							
3								
4	GND							

Software and Firmware

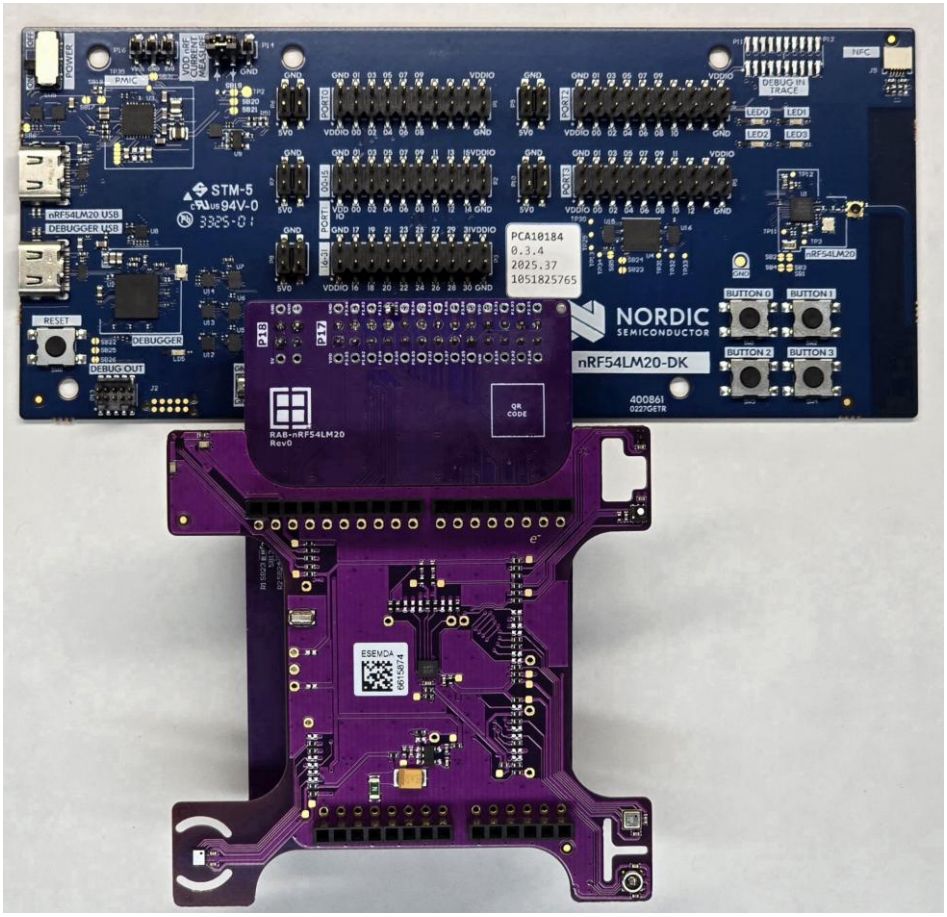
Getting Started with the RAB-nRF54LM20 Adapter Board

The RAB-nRF54LM20 adapter board requires no special configuration before use with any of Rutronik's development boards. The 10K pull-up resistors R1 and R2 are not connected by default; check whether your system requires them for the I2C lines.

1. Mount the RAB-nRF54LM20 adapter to the nRF54LM20-DK Expansion Connectors P17 and P18.



2. Mount the Arduino adapter board to the Arduino Connectors.



3. Use your preferred development environment to start the development or demonstration.

Firmware Examples

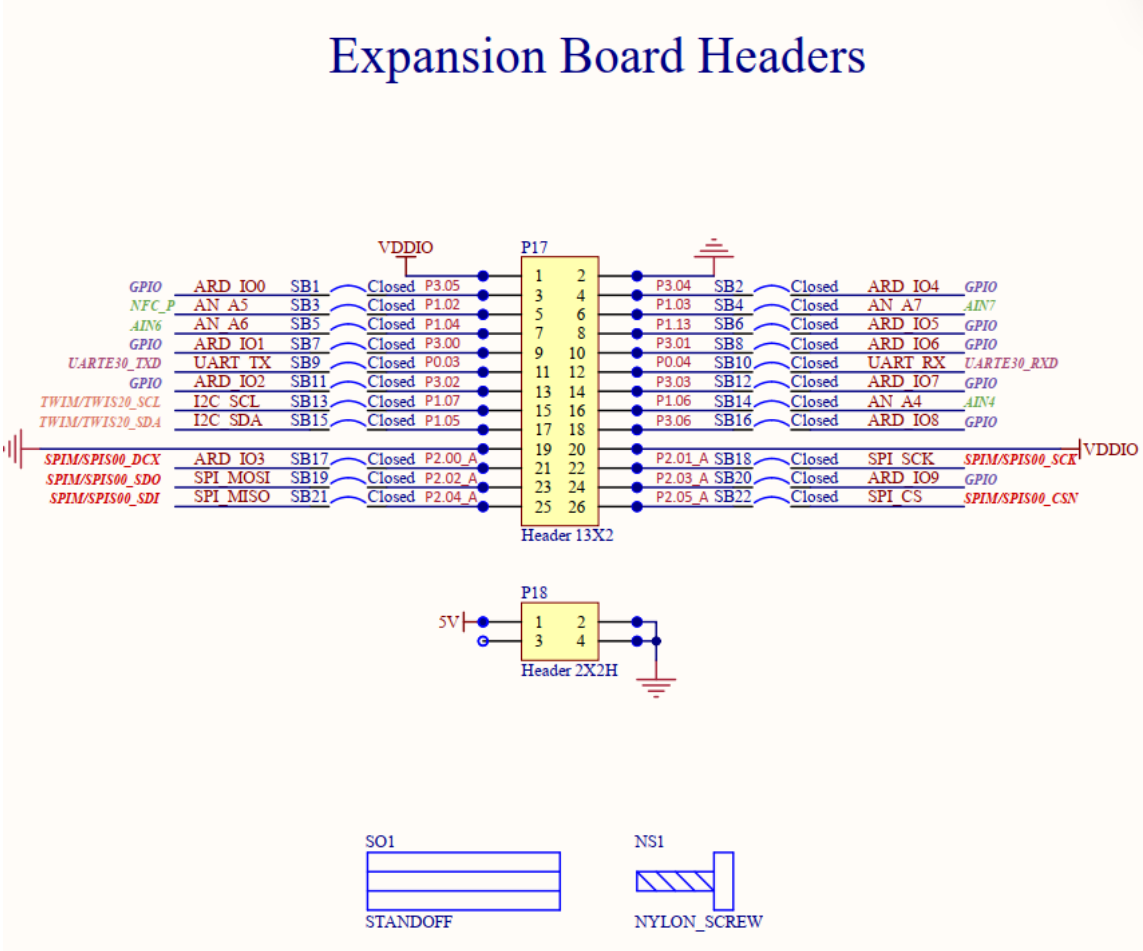
All these examples are on Rutronik System Solutions GitHub and may be imported into Visual Studio Code.

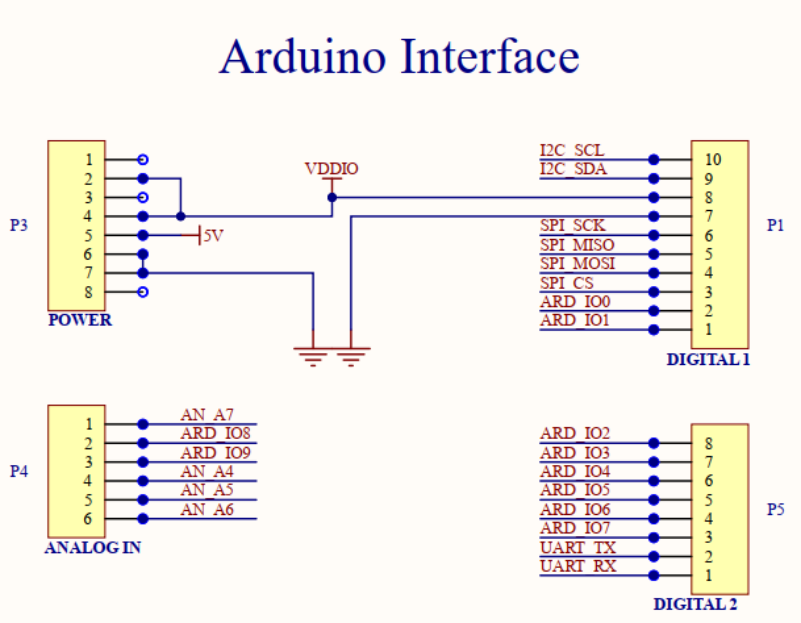
NRF54LM20_DK_RAB7_SensorFusion	Project using nRF54LM20-DK with the RAB7 SensorFusion board.
NRF54LM20_DK_RAB3_Radar	Project using nRF54LM20-DK with the RAB3 Radar board.
NRF54LM20_DK_RAB2_CO2	This project shows how to use the RAB2 CO2 with a Nordic nRF54LM20 DK.

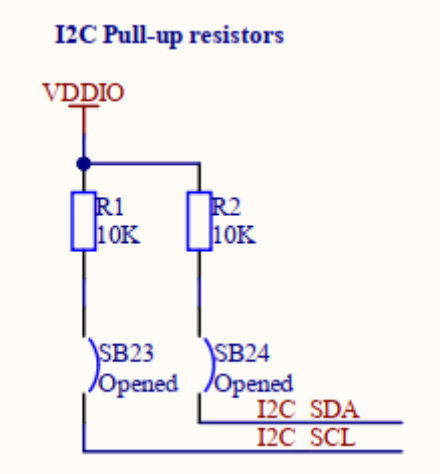
Production Data

Schematics

You'll find the schematics of RAB-nRF54LM20 here.







BOM

You'll find the BOM for RAB-nRF54LM20 here.

RAB nRF54LM20 Rev1 BOM

Part Description	Designator	Footprint	Quantity	Value	Specification	Manufacturer	Manufacturer Part No.	Supplier	Supplier part number
M3 Pan Head Machine Screw	NS1	29311	1	P/N29341	M3, 6mm	Keystone Electronics	P/N 29341	Rutronik	BAT4868
Arduino Connector	P1	Arduino_HDR1X10_main	1	75915-310LF	10 pos. 2,54mm	Amphenol ICC	75915-310LF	Rutronik	CONN7354
Arduino Connector	P3, P5	rduino_HDR1X8_main	2	75915-308LF	8 pos. 2,54mm	Amphenol ICC	75915-308LF	Rutronik	CONN8727
Arduino Connector	P4	Arduino_HDR1X6_main	1	75915-306LF	6 pos. 2,54mm	Amphenol ICC	75915-306LF	Rutronik	CONN8726
nRF54 Board Connector	P17	RECEPTACLE_2x13_2.54_5x8.5	1	87606-313LF	13x2 pos. 2,54mm	Amphenol ICC	87606-313LF	Rutronik	CONN11183
nRF54 Board Connector	P18	RECEPTACLE_2x2_2.54_5x8.5	1	87606-302LF	2X2 pos. 2,54mm	Amphenol ICC	87606-302LF	Rutronik	CONN9239
Thick Film Resistor	R1, R2	R0402	2	10K	5% 0,063W	Yageo	RC0402JR-1310KL	Rutronik	WRC51147
0 Ohms Jumper	SB1, SB2, SB3, SB4, SB5, SB6, SB7, SB8, SB9, SB10, SB11, SB12, SB13, SB14, SB15, SB16, SB17, SB18, SB19, SB20, SB21, SB22	SB_0402	22	0R	5% 0,063W	Yageo	RC0402FR-070RL	Rutronik	WRC53079
5mm Hex Female Standoff	SO1	24475	1	P/N25512	M3, 15mm	Keystone Electronics	P/N 25512	Rutronik	BAT6084

Mechanical Layout

You'll find the Drawing for RAB-nRF54LM20 here.

